

DEMOGRAPHIC INFORMATION

PATIENT

NAME	
DATE OF BIRTH	
SEX	M
MEDICAL RECORD #	EP03613952
ACCESSION #	
LAB NUMBER	
FAMILY NUMBER	2399218

TEST INFORMATION

TEST NAME	Rapid Duo WGS
TEST CODE	1823
SAMPLE TYPE	
DATE COLLECTED	
DATE RECEIVED	06/26/2025
DATE REPORTED	09/02/2025

RECIPIENT

PHYSICIAN NAME	ELYSSA LEWIS
FACILITY	NCGC
LOCATION	LOUISVILLE KY 40202
PHONE	502-588-0850
FAX	502-588-0861

CLINICAL INDICATION

Based on the submitted clinical information, the patient has intrauterine growth retardation, placenta previa, nuchal cord, abnormal brain morphology, abnormal reflex, brisk reflexes, clonus, hypertonia, poor suck, supraventricular tachycardia, arrhythmia, respiratory insufficiency, hyponatremia, neonatal hyperbilirubinemia, and sacral dimple.

We have also received a sample from the mother (Lab#3454525) of this individual.

RESULTS



INDETERMINATE
FINDINGS

DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
SCN5A-Related Cardiac Diseases	Autosomal Dominant	SCN5A: c.3947G>A, p.R1316Q	Sequence Variant	Heterozygous	Mother	Variant Of Uncertain Significance
CAMK2D- Associated Neurodevelopmenta l Disorder With Or Without Dilated Cardiomyopathy	Autosomal Dominant	CAMK2D: c.946+374A>G	Sequence Variant	Heterozygous	Unknown (Mother Is Negative)	Variant Of Uncertain Significance

NAME: [REDACTED]
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SEX: [REDACTED]

RESULTS SUMMARY

Two heterozygous variants of uncertain significance respectively in the SCN5A and CAMK2D genes were detected, possibly related to this patient's phenotype.

CAMK2D:c.946+374A>G is an eligible variant for RNAseq. The sample received for testing will be used, and an addendum report will be issued upon completion of RNAseq.

Multiple regions of absence of heterozygosity larger than 5 Mb were detected. The total amount of AOH is around 220 Mb, suggestive of consanguinity.

chr1: 125157781-189277911
chr1: 87712482-125079667
chr4: 14804723-23186485
chr5: 107377917-124511107
chr5: 151187231-172706022
chr8: 124272118-132565341
chr9: 203802-8261910
chr9: 32735964-38773523
chr9: 68369456-97552458
chr11: 196944-11339249
chr13: 105971080-114271538

ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)

✓ None Detected.

Other Incidental Findings

✓ None Detected.

RECOMMENDATIONS

- It is recommended that correlation of these findings with the clinical phenotype be performed.
- Genetic counseling is recommended.
- For sequence variants, testing is available for this individual's at risk relatives under test code 1580. Visit <https://www.baylorgenetics.com/known-familial-mutations-policy/> for further information. Please consult with the lab for available test codes for other types of variants (CNV, STR, mito).
- Contact laboratory for requests regarding reanalysis and raw data.

NAME: [REDACTED]
DATE OF BIRTH: [REDACTED]
SEX: [REDACTED]

INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
SCN5A-Related Cardiac Diseases	Autosomal Dominant	SCN5A: c.3947G>A, p.R1316Q	Sequence Variant	Heterozygous	Mother	Variant Of Uncertain Significance

GENE DESCRIPTION

The SCN5A gene encodes an integral membrane protein, which functions as a tetrodotoxin-resistant voltage-gated sodium channel subunit. This protein is critical for the fast upstroke of the cardiac action potential in the sodium current.

DISEASE DESCRIPTION

Defects in the SCN5A gene may cause a spectrum of cardiac disorders due to loss-of-function disease-causing variants, including autosomal dominant Brugada syndrome 1 (OMIM: 601144), dilated cardiomyopathy 1E (OMIM: 601154), non-progressive or progressive heart block IA (OMIM: 113900), familial atrial fibrillation 10 (OMIM: 614022), and autosomal recessive sick sinus syndrome 1 (OMIM: 608567). Brugada syndrome 1 is characterized by an ST segment elevation in the right precordial electrocardiogram leads (so-called type 1 ECG) and a high incidence of sudden death with structurally normal hearts. Dilated cardiomyopathy 1E is characterized by ventricular dilation and impaired systolic function, resulting in congestive heart failure and arrhythmias. Non-progressive or progressive heart block IA are characterized by progressive or non-progressive alteration of cardiac conduction through the His-Purkinje system with right or left bundle branch block and widening of QRS complexes, leading to complete atrioventricular block and causing syncope and sudden death. Familial atrial fibrillation 10 is characterized by episodes of rapid/irregular heart beats, atrial fibrillation, syncope, or sudden death without demonstrable cardiac or noncardiac causes to account for the episode. Sick Sinus syndrome 1 is characterized by atrial tachycardia coexisting with sinus bradycardia with electrocardiogram typically showing sinus bradycardia, sinus arrest, and/or sinoatrial block. In addition, defects in the SCN5A gene may also cause Long QT syndrome 3 (OMIM: 603830), due to gain-of-function disease-causing variants, which is characterized by a prolonged QT interval and polymorphic ventricular arrhythmias (torsade de pointes), leading to recurrent syncope, seizures, or sudden death. Reduced penetrance and variable expressivity and age of onset in SCN5A-related cardiac diseases have been demonstrated.

VARIANT DETAILS - SCN5A: c.3947G>A, p.R1316Q

- (GRCh38)chr3:38562431C>T - NM_198056.3
- The c.3947G>A (p.R1316Q) variant in the SCN5A gene has been described as uncertain significance in ClinVar (ID: 647822). This variant has been previously reported in a heterozygous state in an individual with sudden infant death syndrome (PMID: 32789579).
- This variant has been observed with a frequency of < 0.005% in the gnomAD v4 dataset.
- This missense variant occurs at a residue with high evolutionary conservation, and its effect is predicted to be inconclusive (REVEL = 0.868, AlphaMissense = 0.526, CADD = 28.5).

NAME: [REDACTED]
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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
CAMK2D-Associated Neurodevelopmental Disorder With Or Without Dilated Cardiomyopathy	Autosomal Dominant	CAMK2D: c.946+374A>G	Sequence Variant	Heterozygous	Unknown (Mother Is Negative)	Variant Of Uncertain Significance

GENE DESCRIPTION

The CAMK2D gene encodes the delta subunit of a calcium/calmodulin-dependent serine/threonine protein kinase. Calcium/calmodulin dependent signaling is involved in key regulatory pathways including neuronal synaptic signaling and cardiac cell function.

DISEASE DESCRIPTION

Defects in the CAMK2D gene may cause autosomal dominant CAMK2D-associated neurodevelopmental disorder with or without dilated cardiomyopathy (PMID: 38272033), which is characterized by developmental delay, intellectual disability, structural brain anomalies, ocular anomalies, dysmorphic facial features, abnormal muscle tone, dilated cardiomyopathy, musculoskeletal anomalies, gastrointestinal problems, and behavioral abnormalities. Less common clinical features may include seizures and genitourinary anomalies. Genotype-phenotype correlations show that gain-of-function variants in the CAMK2D gene appear to cause both neurological problems and dilated cardiomyopathy, while loss-of-function variants may only lead to neurodevelopmental defects. Currently, the gene-disease association between CAMK2D and the associated neurodevelopmental disorder with or without dilated cardiomyopathy is supported by a limited number of affected individuals. Further clinical and genetic studies are warranted to better define disease onset, penetrance, and genotype-phenotype correlations.

VARIANT DETAILS - CAMK2D: c.946+374A>G

- (GRCh38)chr4:113512954T>C - NM_172115.3
- The c.946+374A>G variant in the CAMK2D gene has not been described in ClinVar.
- This variant has not been observed in the gnomAD v4 dataset.
- This intronic variant is predicted to possibly create a cryptic splice donor site (SpliceAI: 0.33). The functional impact of this change is uncertain.

NAME: [REDACTED]
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QUALITY METRICS

Percentage depth of coverage at 20x: **99.72%**
Mean depth of coverage: **58.18x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATXN8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTBD9, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBF1, TGFBF2, TMEM127, TMEM43, TNNC1, TNNI3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

NAME: [REDACTED]
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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

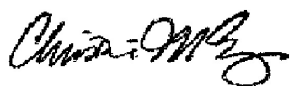
REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

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